SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – Industry based problem solving

Course Code:	**DSA0404**	**Course Title:**	**FUNDAMENTALS OF DATA SCIENCE**
CO Assessed:	**CO4 –Industry based problem solving**	**Assessment:**	**Assessment Tool 2– Industry based problem solving**
Weightage:	**40% of CIA**	**Total Marks:**	**100**
CO4 Statement:	To apply the statistical inferences such as testing of hypothesis and point estimates for the given data		
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Code of Conduct

I Deva Dharshini(192472350) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Deva Dharshini

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	

1.A cybersecurity system historically detects 95% of malicious activities. After deploying a new detection algorithm, it correctly detects 194 out of 200 attacks. Test whether the new algorithm has significantly improved the detection rate using a p-value. Problem

A cybersecurity system historically detects 95% of malicious activities. After deploying a new detection algorithm, it correctly detects 194 out of 200 attacks. Test whether the new algorithm has significantly improved the detection rate using a p-value.

1. Identify the Population and Sample

The population consists of all malicious activities or cyberattacks that the cybersecurity system may encounter in real-world operation.

The sample consists of 200 randomly selected attacks tested using the new detection algorithm.

Given:

- Sample size, $n = 200$
- Correctly detected attacks, $x = 194$
- Undetected attacks = $200 - 194 = 6$
- Historical detection rate = 95%

2. Identify the Parameter of Interest

The parameter of interest is the true population detection rate of the new cybersecurity algorithm.

Let p represent the proportion of all malicious activities that are correctly detected by the new algorithm.

The historical detection rate is:

$p_0 = 0.95$

The objective is to determine whether the new algorithm has improved the detection rate beyond 95%.

3. Calculate the Point Estimate

The point estimate for the population detection rate is the sample proportion.

$\hat{p} = x/n$

$\hat{p} = 194/200$

$\hat{p} = 0.97$

Therefore:

$\hat{p} = 97\%$

The new algorithm detected 97% of the attacks in the sample.

The observed improvement is:

97% − 95% = 2 percentage points.

Thus, although the sample shows a 2-percentage-point improvement, a hypothesis test is required to determine whether this improvement is statistically significant.

4. State the Null Hypothesis H_0

The historical detection rate is 95%.

The null hypothesis states that the new algorithm has not improved the detection rate.

$H_0: p = 0.95$

5. State the Alternative Hypothesis H_1

The purpose of the test is to determine whether the new algorithm has improved the detection rate.

Therefore:

$H_1: p > 0.95$

This is a right-tailed test.

6. Select the Significance Level α

The problem does not specify a significance level. Therefore, the standard significance level of 5% is used.

$\alpha = 0.05$

7. Calculate the Test Statistic

Since this is a hypothesis test for a population proportion, the one-sample proportion z-test is used.

The test statistic is:

$z = (\hat{p} - p_0) / \sqrt{[p_0(1 - p_0)/n]}$

Substituting the values:

$z = (0.97 - 0.95) / \sqrt{[(0.95)(0.05)/200]}$

$z = 0.02 / \sqrt{(0.0475/200)}$

$z = 0.02 / \sqrt{0.0002375}$

$z = 0.02 / 0.01541$

$z \approx 1.30$

Therefore:

Test statistic = 1.30

8. Calculate the p-value

Because the alternative hypothesis is:

$H_1: p > 0.95$

this is a right-tailed test.

Therefore:

p-value = $P(Z > 1.30)$

From the standard normal distribution:

p-value ≈ 0.097

9. Compare p-value with α

The values are:

p-value = 0.097

$\alpha = 0.05$

Comparison:

$0.097 > 0.05$

Since the p-value is greater than α, we fail to reject H_0.

10. Industry-Oriented Conclusion

The new cybersecurity algorithm achieved a sample detection rate of 97%, which is higher than the historical detection rate of 95%.

However, the p-value is approximately 0.097, which is greater than the 5% significance level.

Therefore, there is insufficient statistical evidence to conclude that the new algorithm has significantly improved the malicious-activity detection rate.

From a cybersecurity industry perspective, the observed increase from 95% to 97% is encouraging, but it cannot be considered statistically significant based on the current sample of 200 attacks. The organization should conduct additional testing using a larger and more diverse set of cyberattacks before claiming that the new algorithm provides a significant improvement.

Final decision:

Reject H_0? No.

Conclusion: The new algorithm has not demonstrated a statistically significant improvement at the 5% significance level.

2. A hospital introduces a new appointment scheduling system. Before implementation, the average patient waiting time was 42 minutes. After implementation, a sample of 64 patients has an average waiting time of 38 minutes with standard deviation 12 minutes. Test whether the new system significantly reduces waiting time using the p-value approach.Problem

A hospital introduces a new appointment scheduling system. Before implementation, the average patient waiting time was 42 minutes. After implementation, a sample of 64 patients has an average waiting time of 38 minutes with a standard deviation of 12 minutes. Test whether the new system significantly reduces waiting time using the p-value approach.

1. Identify the Population and Sample

The population consists of all patients using the hospital's appointment scheduling system.

The sample consists of 64 patients whose waiting times were measured after implementation of the new system.

Given:

- $n = 64$
- Sample mean, $\bar{x} = 38$ minutes
- Sample standard deviation, $s = 12$ minutes
- Historical mean, $\mu_0 = 42$ minutes

2. Identify the Parameter of Interest

The parameter of interest is the true average waiting time of all patients after implementation of the new appointment scheduling system.

Let μ represent the population mean waiting time.

The historical average is:

$\mu_0 = 42$ minutes.

The objective is to determine whether μ is significantly less than 42 minutes.

3. Calculate the Point Estimate

The point estimate of the population mean is the sample mean.

$\bar{x} = 38$ minutes

Therefore, the estimated average waiting time after implementing the new system is 38 minutes.

The observed reduction is:

$42 - 38 = 4$ minutes.

Thus, the new system appears to reduce average waiting time by 4 minutes.

4. State the Null Hypothesis H_0

The null hypothesis states that the new appointment system has not reduced the average waiting time.

H_0: $\mu = 42$ minutes

Equivalently, for the purpose of the reduction test:

H_0: $\mu \geq 42$ minutes

5. State the Alternative Hypothesis H_1

The hospital wants to determine whether the new system reduces waiting time.

Therefore:

H_1: $\mu < 42$ minutes

This is a left-tailed test.

6. Select the Significance Level α

The problem does not specify α, so the standard 5% significance level is selected.

$\alpha = 0.05$

7. Calculate the Test Statistic

The population standard deviation is not given. Since the sample standard deviation is used, a one-sample t-test is appropriate.

The test statistic is:

$t = (\bar{x} - \mu_0) / (s/\sqrt{n})$

Substituting:

$t = (38 - 42) / (12/\sqrt{64})$

$t = -4 / (12/8)$

$t = -4/1.5$

$t \approx -2.67$

Degrees of freedom:

$df = n - 1$

$df = 64 - 1 = 63$

Therefore:

$t \approx -2.67$

8. Calculate the p-value

Since this is a left-tailed test:

p-value = $P(T_{63} < -2.67)$

The approximate p-value is:

p-value ≈ 0.0048

9. Compare p-value with α

p-value = 0.0048

$\alpha = 0.05$

Therefore:

$0.0048 < 0.05$

Since the p-value is less than α, we reject H_0.

10. Industry-Oriented Conclusion

There is sufficient statistical evidence at the 5% significance level to conclude that the new appointment scheduling system significantly reduces the average patient waiting time.

The sample average waiting time decreased from 42 minutes to 38 minutes, representing a reduction of 4 minutes.

The p-value of approximately 0.0048 is much smaller than 0.05, which indicates that the observed reduction is unlikely to be due to random sampling variation alone.

As an additional interpretation, the approximate 95% confidence interval for the population mean is:

$\bar{x} \pm t_{0.975,63}(s/\sqrt{n})$

Using $t \approx 2.00$:

$38 \pm 2(12/8)$

38 ± 3

Therefore:

95% CI $\approx (35, 41)$ minutes.

The historical mean of 42 minutes is outside this interval, providing additional evidence that the average waiting time has decreased.

From a healthcare industry perspective, the result supports implementation of the new scheduling system. Reduced waiting time can improve patient satisfaction, decrease overcrowding, improve appointment management, and increase the efficiency of hospital resources.

Final decision:

Reject H_0.

Conclusion: The new appointment scheduling system has significantly reduced average patient waiting time.

3. A social media platform claims that the average daily usage time of its users is 3 hours. A random sample of 100 users has an average usage time of 3.2 hours with standard deviation 0.8 hours. Determine whether the data provide sufficient evidence to reject the company claim at $\alpha = 0.05$.

Problem

A social media platform claims that the average daily usage time of its users is 3 hours. A random sample of 100 users has an average usage time of 3.2 hours with a standard deviation of 0.8 hours. Determine whether the data provide sufficient evidence to reject the company's claim at $\alpha = 0.05$.

1. Identify the Population and Sample

The population consists of all users of the social media platform.

The sample consists of 100 randomly selected users whose daily usage times were recorded.

Given:

- $n = 100$
- $\bar{x} = 3.2$ hours
- $s = 0.8$ hours
- Claimed mean = 3 hours

2. Identify the Parameter of Interest

The parameter of interest is the true average daily usage time of all users.

Let μ represent the population mean daily usage time.

The company's claim is:

$\mu = 3$ hours.

3. Calculate the Point Estimate

The point estimate of the population mean is the sample mean:

$\bar{x} = 3.2$ hours

Therefore, the estimated average daily usage time is 3.2 hours.

The difference between the sample estimate and the company's claim is:

$3.2 - 3.0 = 0.2$ hours

Since 0.2 hours = 12 minutes, the sample suggests that users spend approximately 12 additional minutes per day on the platform compared with the claimed average.

4. State the Null Hypothesis H_0

The company's claim is that the average usage time is exactly 3 hours.

H_0: $\mu = 3$ hours

5. State the Alternative Hypothesis H_1

Since the question asks whether there is sufficient evidence to reject the company's claim, we test whether the true average is different from 3 hours.

H_1: $\mu \neq 3$ hours

This is a two-tailed test.

6. Select the Significance Level α

The problem directly provides:

$\alpha = 0.05$

Therefore, the test is conducted at the 5% significance level.

7. Calculate the Test Statistic

Since the population standard deviation is unknown and the sample standard deviation is given, a one-sample t-test is appropriate.

The test statistic is:

$t = (\bar{x} - \mu_0)/(s/\sqrt{n})$

Substituting:

$t = (3.2 - 3.0)/(0.8/\sqrt{100})$

$t = 0.2/(0.8/10)$

$t = 0.2/0.08$

$t = 2.50$

Degrees of freedom:

$df = 100 - 1 = 99$

Therefore:

$t = 2.50$, $df = 99$

8. Calculate the p-value

Because this is a two-tailed test:

p-value $= 2P(T_{99} > 2.50)$

The approximate p-value is:

p-value ≈ 0.014

9. Compare p-value with α

p-value ≈ 0.014

$\alpha = 0.05$

Therefore:

$0.014 < 0.05$

Since the p-value is less than α, we reject H_0.

10. Industry-Oriented Conclusion

There is sufficient statistical evidence at the 5% significance level to reject the social media company's claim that the average daily usage time is 3 hours.

The sample mean is 3.2 hours, which is 0.2 hours or 12 minutes higher than the claimed value.

The p-value of approximately 0.014 is less than 0.05, indicating that the difference is statistically significant.

A 95% confidence interval can also be calculated approximately as:

$\bar{x} \pm t_{0.975,99}(s/\sqrt{n})$

Using $t \approx 1.984$:

$3.2 \pm 1.984(0.8/10)$

3.2 ± 0.159

Therefore:

95% CI ≈ (3.041, 3.359) hours.

The claimed value of 3 hours is not included in this interval.

From a social media analytics perspective, this result indicates that actual average user engagement is likely higher than the company's stated 3-hour figure. This information can be useful for advertising planning, server-capacity management, content recommendation systems, user engagement analysis, and business forecasting.

Final decision:

Reject H_0.

Conclusion: There is sufficient evidence to reject the company's claim that the average daily usage time is 3 hours.

4. A data science team develops a new classification model. The existing model has an accuracy of 82%, while the new model achieves 86% accuracy on a randomly selected test sample of 500 observations. Formulate suitable null and alternative hypotheses and determine whether the improvement is statistically

Problem

A data science team develops a new classification model. The existing model has an accuracy of 82%, while the new model achieves 86% accuracy on a randomly selected test sample of 500 observations. Formulate suitable null and alternative hypotheses and determine whether the improvement is statistically significant using a p-value.

1. Identify the Population and Sample

The population consists of all observations on which the classification model may be applied under similar operating conditions.

The sample consists of 500 randomly selected test observations evaluated using the new classification model.

Given:

- $n = 500$
- Existing model accuracy = 82%
- New model accuracy = 86%

Number of correctly classified observations:

$x = 0.86 \times 500$

$x = 430$

Therefore, the new model correctly classifies approximately 430 of the 500 test observations.

2. Identify the Parameter of Interest

The parameter of interest is the true classification accuracy of the new model.

Let p represent the true population accuracy of the new classification model.

The existing model provides a benchmark accuracy of:

$p_0 = 0.82$

The objective is to determine whether the new model's true accuracy is greater than 82%.

3. Calculate the Point Estimate

The sample proportion is:

$\hat{p} = x/n$

$\hat{p} = 430/500$

$\hat{p} = 0.86$

Therefore:

$\hat{p} = 86\%$

The observed improvement is:

86% − 82% = 4 percentage points.

4. State the Null Hypothesis H_0

The null hypothesis states that the new model has not improved the accuracy compared with the existing benchmark.

H_0: $p = 0.82$

5. State the Alternative Hypothesis H_1

The data science team wants to determine whether the new model has improved accuracy.

Therefore:

H_1: $p > 0.82$

This is a right-tailed test.

6. Select the Significance Level α

Since α is not specified, use the standard:

$\alpha = 0.05$

7. Calculate the Test Statistic

For a one-sample proportion test:

$z = (\hat{p} - p_0)/\sqrt{[p_0(1 - p_0)/n]}$

Substituting:

$z = (0.86 - 0.82)/\sqrt{[(0.82)(0.18)/500]}$

$z = 0.04/\sqrt{(0.1476/500)}$

$z = 0.04/\sqrt{0.0002952}$

$z = 0.04/0.01718$

$z \approx 2.33$

Therefore:

Test statistic ≈ 2.33

8. Calculate the p-value

Since this is a right-tailed test:

p-value $= P(Z > 2.33)$

From the standard normal distribution:

p-value ≈ 0.010

9. Compare p-value with α

p-value ≈ 0.010

$\alpha = 0.05$

Therefore:

$0.010 < 0.05$

Since the p-value is smaller than α, we reject H_0.

10. Industry-Oriented Conclusion

There is sufficient statistical evidence at the 5% significance level to conclude that the new classification model has a higher accuracy than the existing model's 82% benchmark.

The new model achieved an accuracy of 86%, representing an observed improvement of 4 percentage points.

The p-value of approximately 0.010 is less than 0.05. Therefore, the improvement is statistically significant.

From a machine-learning industry perspective, this result supports further consideration of the new model for deployment. However, model accuracy should not be the only criterion. The data science team should also examine precision, recall, F1-score, confusion matrix results, computational requirements, inference speed, robustness, and performance on independent datasets.

A particularly important point is that this test compares the new model's accuracy against the existing model's historical accuracy of 82%. If both models were tested on the same observations, a paired comparison or an appropriate two-model statistical test may be preferable.

Final decision:

Reject H_0.

Conclusion: Based on the given benchmark and sample information, the new model's improvement from 82% to 86% is statistically significant at the 5% significance level.

5. A manufacturer claims that its production process has a defect rate of only 2%. From 2,000 randomly selected products, 55 are defective. Perform a hypothesis test at the 5% significance level using both (a) confidence interval and (b) p-value approaches. Compare the conclusions.

A manufacturer claims that its production process has a defect rate of only 2%. From 2,000 randomly selected products, 55 are defective. Perform a hypothesis test at the 5% significance level using both:

p-value approach

Confidence control approach

Compare the conclusions.

1. Identify the Population and Sample

The population consists of all products manufactured by the production process.

The sample consists of 2,000 randomly selected products that were inspected for defects.

Given:

- $n = 2{,}000$
- Number of defective products = 55
- Claimed defect rate = 2%
- $\alpha = 0.05$

2. Identify the Parameter of Interest

The parameter of interest is the true population defect rate.

Let p represent the actual proportion of defective products produced by the manufacturing process.

The manufacturer's claim is:

$p_0 = 0.02$

The objective is to determine whether the actual defect rate is consistent with the claimed 2%.

3. Calculate the Point Estimate

The sample defect rate is:

$\hat{p} = x/n$

$\hat{p} = 55/2000$

$\hat{p} = 0.0275$

Therefore:

$\hat{p} = 2.75\%$

The sample estimate of the defect rate is 2.75%.

The difference between the sample defect rate and the claimed rate is:

2.75% − 2% = 0.75 percentage points.

Therefore, the observed sample defect rate is higher than the manufacturer's claimed rate.

4. State the Null Hypothesis H_0

The manufacturer's claim is that the defect rate is 2%.

Therefore:

$H_0: p = 0.02$

5. State the Alternative Hypothesis H_1

To test whether the claim is incorrect:

$H_1: p \neq 0.02$

This is a two-tailed test.

If the specific industrial concern is whether the process is worse than claimed, the practical direction is $p > 0.02$. However, for testing the stated equality claim using the confidence interval and p-value approach, the two-sided hypothesis is appropriate.

6. Select the Significance Level α

The problem specifies:

$\alpha = 0.05$

Therefore, the test is performed at the 5% significance level.

7. Calculate the Confidence Interval and Test Statistic

A. 95% Confidence Interval

For a population proportion, the approximate 95% confidence interval is:

$\hat{p} \pm 1.96\sqrt{[\hat{p}(1 - \hat{p})/n]}$

Substituting:

$\hat{p} = 0.0275$

$n = 2000$

Standard error:

$SE = \sqrt{[(0.0275)(0.9725)/2000]}$

$SE \approx 0.003657$

Margin of error:

$ME = 1.96 \times 0.003657$

$ME \approx 0.00717$

Therefore:

$95\%\ CI = 0.0275 \pm 0.00717$

Lower limit:

$0.0275 - 0.00717 = 0.02033$

Upper limit:

$0.0275 + 0.00717 = 0.03467$

Therefore:

$95\%\ CI \approx (0.02033, 0.03467)$

In percentage form:

$95\%\ CI \approx (2.03\%, 3.47\%)$

Interpretation of Confidence Interval

The estimated defect rate is between approximately 2.03% and 3.47% at the 95% confidence level.

The manufacturer's claimed value of 2% is slightly below the lower confidence limit of approximately 2.03%.

Therefore, the value 2% is not contained in the approximate 95% confidence interval.

This provides evidence against the manufacturer's claim at the 5% significance level.

B. Test Statistic

For the one-sample proportion test:

$z = (\hat{p} - p_0)/\sqrt{[p_0(1 - p_0)/n]}$

Substituting:

$z = (0.0275 - 0.02)/\sqrt{[(0.02)(0.98)/2000]}$

$z = 0.0075/\sqrt{(0.0196/2000)}$

$z = 0.0075/0.00313$

$z \approx 2.40$

Therefore:

$z \approx 2.40$

8. Calculate the p-value

Because the alternative hypothesis is:

$H_1: p \neq 0.02$

this is a two-tailed test.

Therefore:

p-value = $2P(Z > 2.40)$

From the standard normal distribution:

p-value ≈ 0.016

9. Compare p-value with α

The values are:

p-value ≈ 0.016

$\alpha = 0.05$

Therefore:

$0.016 < 0.05$

Since the p-value is less than α, we reject H_0.

10. Industry-Oriented Conclusion and Comparison

The two statistical approaches provide the same overall decision.

Confidence Interval Approach

The 95% confidence interval is approximately:

(2.03%, 3.47%)

The claimed defect rate of 2% is not included in the interval.

Therefore, the manufacturer's claim is not supported by the confidence interval.

p-value Approach

The p-value is approximately:

0.016

Since:

$0.016 < 0.05$

we reject H_0.

Therefore, the p-value approach also provides sufficient evidence against the manufacturer's claim.

Comparison of Both Methods

Method	**Result**	**Decision**
95% Confidence Interval	2% is outside (2.03%, 3.47%)	Reject H_0
p-value	$0.016 < 0.05$	Reject H_0

Both approaches lead to the same conclusion.

The sample defect rate is 2.75%, which is higher than the claimed 2%. The statistical analysis indicates that this difference is significant at the 5% level.

From a manufacturing and quality-control perspective, the company should not continue to assume that its process has a defect rate of only 2%. The observed defect rate is higher, and the statistical evidence indicates that the difference is unlikely to be caused solely by random sampling variation.

The manufacturer should investigate possible causes of defects, including machine calibration, raw-material quality, production conditions, operator performance, maintenance procedures, and inspection processes.

A defect rate of 2.75% means approximately 27.5 defective products per 1,000 products, whereas a 2% defect rate corresponds to 20 defective products per 1,000 products. Although the difference may appear small, it can result in a substantial number of additional defective products when production volumes are large.

Final Conclusion

The sample estimate of the defect rate is 2.75%.

The approximate 95% confidence interval is:

(2.03%, 3.47%)

The test statistic is:

$z \approx 2.40$

The p-value is:

p-value ≈ 0.016

Since:

$0.016 < 0.05$

the null hypothesis is rejected.

Therefore, there is sufficient statistical evidence at the 5% significance level to reject the manufacturer's claim that the production defect rate is 2%.

Both the confidence interval and p-value approaches produce the same conclusion: the actual defect rate appears to be higher than the claimed 2%, and the manufacturing process should be investigated and improved.